

Multi-Wavelength Diagnostics of Pre-Flare Evolution with Aditya-L1: From Chromosphere to Corona

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ABSTRACT

Exploring the pre-flare phase using Aditya L1 data

Key words: keyword1 – keyword2 – keyword3

1 INTRODUCTION

Flare is intense, major energy comes from the chromosphere NUV, we can see the chromosphere, suit got high cadence along with hard X-ray and soft X-ray uv observations; kusano and bamba works, Lucia kent and Brandon observed wing enhancement in IRIS (Mg triplet). x-ray observations; Ananth soft X-ray cubesat - abundance change2023,2025, D.Silva-statistical studies 2023, Battagila2023

pre-flare phase time using spectral signatures from IRIS and EIS

In this Solar Ultraviolet Imaging Telescope (SUIT; D. Tripathi et al. 2017; D. Tripathi et al. 2024; J. Sarkar et al. 2024) on board Aditya-L1 (D. Tripathi et al. 2023, Sarkar et al. (2025), Tripathi et al. (2017), Tripathi et al. (2025), Tripathi et al. (2023))

In this work, we are studying

- (i) What happens in the pre-flare phase of the solar flare (2-hour window).
- (ii) How Chromosphere (Mg II h) behaves during the pre-flare phase.
- (iii) Is there a hot X-ray onset in all flares?
- (iv) How Aditya-L1 can be a unique mission for observing Solar flare precursors.

2 OBSERVATIONS

In this study, we analyse the pre-flare phase of a solar flare using multiple instruments onboard Aditya-L1, which together provide simultaneous observations of the chromosphere and the corona. The Level-1, Mg II h narrowband Region of Interest (ROI) images ($\approx 491'' \times 491''$) from SUIT/Aditya-L1, with a cadence of approximately 85 s, were used. The data were corrected for scatter and vignetting

For coronal observations, we use data from both SoLEXS (2–22 keV) and HELIOS (CdTe; 10–40 keV), which are high-resolution spectrographs that observe the Sun as a star. We additionally use

Case no.	NOAA no.	Flare class	Flare start time	Flare peak time
Case 4	13738	M1.5	2024-07-10 05:44	2024-07-10 05:59
Case 5	13738	M1.1	2024-07-10 15:25	2024-07-10 15:37
Case 6	13848	X1.8	2024-10-09 01:25	2024-10-09 01:56
Case 7	13878	M1.3	2024-11-01 02:05	2024-11-01 02:16
Case 8	13878	M2.0	2024-11-01 14:18	2024-11-01 14:31
Case 9	13878	M1.0	2024-11-13 00:10	2024-11-13 00:22
Case 10	13878	M1.7	2024-11-13 16:57	2024-11-13 17:08

Table 1. Flare cases being studied, the flare class, start time and peak time are based on the GOES flare catalogue

STIX onboard Solar Orbiter for spectral analysis. We restricted our sample to flares of class M and above, located close to disk centre, and with no major (C, M, or X-class) flare occurring within the preceding two hours.

The selected events that satisfy these criteria are listed in Table 1. For comparison of source locations, we additionally incorporate AIA, HMI observations.

For the comparison of source locations, we also use AIA, HMI, and STIX data.

2.1 Analysis

To capture the overall behaviour of the active region, we plotted the light curves of the SUIT Mg II h 2803 Å filter. The images were further examined for small-scale brightenings using the image-differencing method. These events were then investigated using complementary coronal observations from SoLEXS and HELIOS. To infer the corresponding spatial locations in the corona, we incorporated observations from AIA and HMI.

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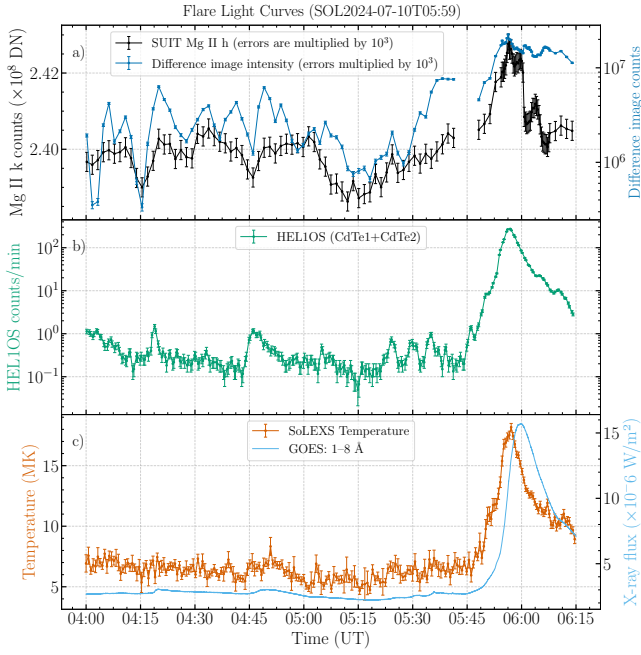


Figure 1. a) Total intensity light curve of the Mg II h ROI images (black). The errors of the individual data points are scaled by 10^3 for visibility. The light curve of the difference-image intensity above the threshold is shown in cyan, with errors scaled by 10^4 for clarity. b) HELIOS light curve (CdTe 1 and CdTe 2, 9–40 keV), binned to 1-minute intervals, c) Temperature derived from SoLEXS (orange) and X-ray flux from the GOES 1–8 Å band (sky blue)

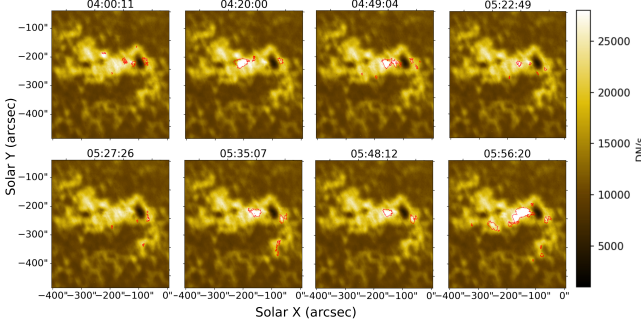


Figure 2. Case 4 : SUIIT time stamp images of pre-flare enhancements

2.2 Flare light curve

All the images are co-aligned using the `mc_coalign` function, publicly available through the `sunkit-image`¹ python package, to remove the satellite drift effect in the image, which will also remove the differential rotation. Further, we selected the common area available in all the frames and plotted the total count of the entire image as a light curve.

2.3 Identifying pre-flare enhancements in SUIIT

To identify the pre-flare enhancements, we adopted the base-difference method. The base image was constructed using the median

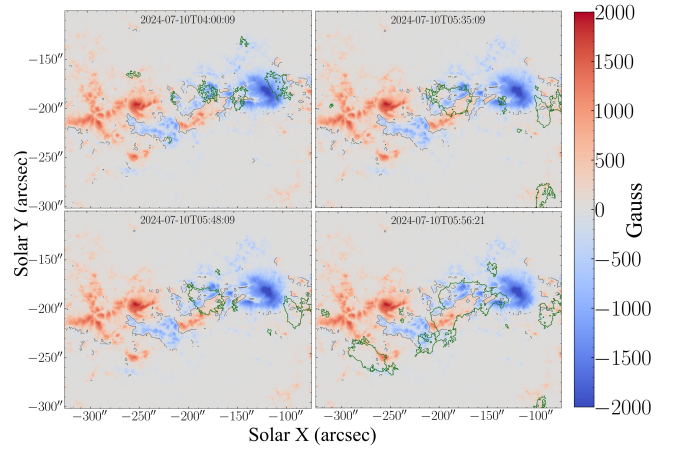


Figure 3. Case 4 HMI images, gray colour line represents PIL (0 Gauss value pixels) and green colour contours are SUIIT enhancement contours from difference image.

of the first five images in the 2-hour pre-flare image sequence, which also minimises first-frame selection bias. The base-difference images contain quiet-Sun and plage-related fluctuations, as well as intensity fluctuations caused by the contamination pattern. To mitigate these effects, we had to set a threshold to see strong enhancements. Mean and standard deviation of the image were affected by bright events. Therefore, we computed the median and standard deviation ($\hat{\sigma}$) from median absolute deviation (MAD) of the difference image, which are less affected by bright events. And we applied the detection threshold given in Eq. 1 to the base-difference image, and contours were drawn around the regions exceeding this threshold.

$$\text{Threshold} = \text{median} + 5\hat{\sigma} \quad (1)$$

$$\hat{\sigma} = k \times \text{MAD} \quad (2)$$

where

$$\tilde{X} = \text{median}(X_i)$$

$$\text{MAD} = \text{median}(|X_i - \tilde{X}|)$$

$$k \approx 1.4826 \text{ (for normal distribution)}$$

Additionally, we imposed a size constraint of 16 pixels (corresponding to a PSF FWHM of ≈ 8 pixels). Only features larger than the PSF were retained. The base-difference total intensity within these selected regions was then used to generate the enhancement light curve.

2.4 HELIOS light curves

To study simultaneous coronal activity, we consider the HELIOS observations, which cover both thermal and non-thermal parts of the flare (10–40 keV). Multiple count enhancement is observed in the 2-hour pre-flare window, but since it is a sun as a star observation, we cannot say the source is the same active region being observed from SUIIT. In most cases, we observe that SUIIT enhancement is with by HELIOS enhancements. These enhancements could be small flares; if these enhancements are flares, we expect some non-thermal emission from them. But we have very few counts; we had to integrate over a few minutes to have enough counts to fit the spectra. However, as we integrate over a large time range, we obtain an averaged spectrum, which is essentially a first-order approximation. We compare

¹ <https://docs.sunpy.org/projects/sunkit-image/en/latest/index.html>

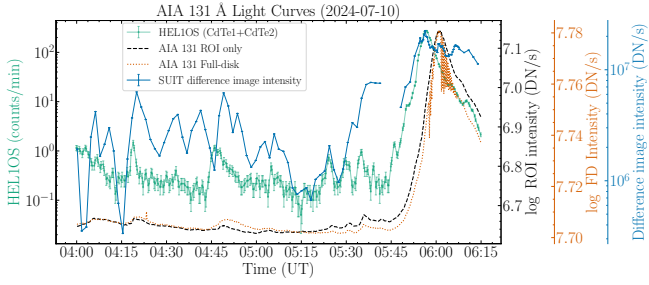


Figure 4. AIA 131 light curves compared with HELIOS light curve

our estimation with STIX for cases 5 to 10. which is described in section 4.

2.5 Temperature from SoLEXS

For the same pre-flare interval, we derived the emission measure and temperature from SoLEXS observations, assuming an isothermal plasma. We had to bin the data for about 2-3 minutes to have enough signal for calculation. SoLEXS temperature follows almost the trend of HELIOS.

2.6 Identifying X-ray Source locations on full disc

Both HELIOS and SoLEXS observations come from the Sun as a star observation. We are not sure of the source location of these small enhancements. Using the AIA observation, try to locate the source of these emissions (at least to the active region). AIA 131 Å and AIA 94 Å observe the hot plasma close to the X-ray in the SDO channels.

We plotted the total counts from the full Sun up to $1.1 R_{\odot}$, the active region of interest (ROI) observed with SUIT, and the HELIOS counts. The full-disc light curve is expected to mimic observations of the Sun as a star in X-rays. We observe a slight delay between HELIOS and AIA 131 light curves, most likely due to a difference in the energy bands of observation. Each enhancement/event in HELIOS can be connected to one active region (see the figure in the appendix, which includes all light curves). This figure indicates which part of the X-ray light curve should be used to infer information about the active region being observed.

2.7 Spectral analysis of pre-flare enhancements

Currently, the HELIOS counts are insufficient to fit the spectra, and with STIX spectra, we are analysing pre-flare enhancements.

2.8 Hot onset cases

In most of the flare cases, we observe elevated temperatures during flare onset in SoLEXS. We also notice HELIOS counts increasing during that period. This appears like a hot onset (previously reported, [Hudson et al. \(2021\)](#), [Hudson \(2025\)](#), [Battaglia et al. \(2023\)](#)). We don't have enough counts in HELIOS to conclude that they are of pure thermal nature. but, we have an STIX observation for the last 6 cases; using STIX data, we analyse the pre-flare enhancements and hot onset part by fitting the thermal and non-thermal parts.

3 RESULTS

3.1 Case 4: M1.5 class flare on July 10 2024

We observe multiple brightenings in the pre-flare time window (Fig. ??). All brightenings occur cospatially with the polarity inversion line (PIL) (A1, A2, A3..), with some located within approximately 10 arcseconds of it. (fig 3). All enhancements are cospatial with AIA 171 images, suggesting their connection with coronal events. Exact timing analysis with other images could not be done due to low cadence in SUIT.

Several of the enhancements exhibit corresponding pre-event signatures in HELIOS, HELIOS counts are not enough to do the spectral analysis. To confirm their source locations, we examined the AIA 131 Å images. The continuous enhancement began to appear at the flaring location after 05:28, which is approximately 31 minutes before the flare's soft X-ray peak time (based on the GOES data). We can observe multiple small events in the same active region (confirmed by AIA 131 plot) in this period. The coronal temperature measured from SoLEXS reaches approximately 6 MK and shows a rapid increase after 05:46, which is about 15 minutes before the soft X-ray peak.

3.2 Case 5: M1.1 class flare on July 10 2024

In this case as well, multiple brightenings are observed during the pre-flare phase. Four major brightening sites were identified, including the flare locations (B1, B2, B3, and B4). All brightenings are on or close to the PIL (Cannot differentiate the exact location due to the PSF size).

After 14:45 onwards continuous brightening is observed, B2 and B3 are constant contributors and B1 is pulsating, note that all three regions are connected (see 171 images in fig). SoLEXS temperature does not show much variation, it lies between 6-7 MK.

(add iris images)

3.3 Case 6: X1.8 class flare on October 9 2024

It is the strong flares in our dataset and it was a sigmoid flare (sigmoid flare ref). Minor pre-flare activity was detected in this active region. A small brightening was observed at point C1.

3.4 Case 7: M1.3 class flare on November 1 2024

Similar to the other cases, we also observe pre-flare brightenings in this case. All significant brightenings originate around the flaring location—specifically those for which HELIOS shows enhancements above the error bars. These pre-flare brightenings occur on and close to the polarity inversion line (PIL), whereas the main flare takes place exactly above the PIL.

Using AIA source-localisation techniques, we found that the X-ray peaks correspond to the same region identified in the SUIT observations. STIX spectra for this event show three significant peaks, all exhibiting non-thermal emission, indicating that these correspond to small flare events.

In this case, we do not observe a single continuously enhanced region; instead, the emission is pulsating, consistent with a series of small flares.

3.5 Case 8: M2.0 class flare on November 1 2024

Only one enhancement was observed in the pre-flare phase, on the flare location. Pre-flare enhancement and flare are happening slightly away from PIL. STIX spectral fit suggest that pre-flare enhancement happening at 13:13 is also a small flare. Continuous enhancement started at the flare location and the loop foot point about 30 min earlier than the flare peak.

3.6 Case 9: M1.0 class flare on November 13 2024

Only two brightening was observed in pre-flare phase where one among was on the flare location. STIX spectra confirm that small enhancement is the flare. Small continuous enhancement was observed before flare on the flaring location and it stopped just before the flare.

3.7 Case 10: M1.7 class flare on November 13 2024

One small enhancement was observed but actual flare was happened another location but still this active region responded, suggesting larger connectivity between active region.

Our pre-flare study suggests that multiple small brightening events are observed in the chromosphere before the flare. This could be due to a small flare or pure thermal events.

Most of the brightening is happening on the PIL, where a flare is going to take place.

All these small events are counter parts of coronal activity .

We observe that some of these events are not started in the same region but are a response to a small event that happened in another region of the sun. This suggests a larger network connecting all active regions.

Summary table

4 DISCUSSION

Multiple brightenings on the PIL were observed. It can also be a small flare, since it is solar maximum, not to be confused with just an enhancement and thought of as a precursor.

All these multiple small events could lead to destabilising the larger active region and causing a flare.

especially the event happening just before a flare may be the subject of interest that can tell or that can be the signature of a flare.

Further investigation based on HMI could lead to much more concrete evidence for the flare trigger understanding.

5 CONCLUSION

Aditya-L1 can clearly demonstrate the capability of observing pre-flare dynamics in both the chromosphere and the corona by combining both imaging and spectral observations.

Repeating the same procedures during Solar minima could give further confidence in our results.

We need high cadence imaging spectrographs to isolate the source location for better understanding of the solar corona during flares.

ACKNOWLEDGEMENTS

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DATA AVAILABILITY

All data are publicly available on the ISSDC-pradan website.

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Tripathi D., et al., 2025, *Solar Physics*, 300, 30

APPENDIX A: LIST OF PRE-FLARES

APPENDIX B: JUSTIFICATION FOR FEW STEPS

B1 Difference image threshold

APPENDIX C: FLARE LIGHT CURVES

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Case no.	Pre-flare brightening location	Nature of PFB	Hot onset (Yes/No)	Onset temperature	GOES Flare peak time
Case 4	Flaring location	Continuous	Yes	6-7	5:59
Case 5	Flaring location and closeby	Continuous	No	6-7	15:39
Case 6	Flaring location and closeby	Before flare	No	6-7	01:56
Case 7	Flaring location	Pulasting	No	7-8	02:16

Table A1. Case 4 : Pre-flare brightenings timestamps and corresponding intensities from active region 13738.

No.	Timestamp (UT)	Intensity (DN/s $\div 10^6$)	HEL10S (Counts above 12 keV)	STIX	NB08 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-07-10 04:06:16	105.44	-	N/A						
2	2024-07-10 04:09:19	45.82	-							
3	2024-07-10 04:20:00	77.64	19							
4	2024-07-10 04:33:46	61.07	14							
5	2024-07-10 04:41:24	70.40	-							
6	2024-07-10 04:49:04	77.33	-							
7	2024-07-10 04:53:40	65.29	-							
8	2024-07-10 05:02:51	36.37	-							
9	2024-07-10 05:05:55	39.11	-							
10	2024-07-10 05:15:07	13.59	-							
11	2024-07-10 05:24:21	31.26	-							
12	2024-07-10 05:32:02	48.60	-							
13	2024-07-10 05:38:11	88.44	-							

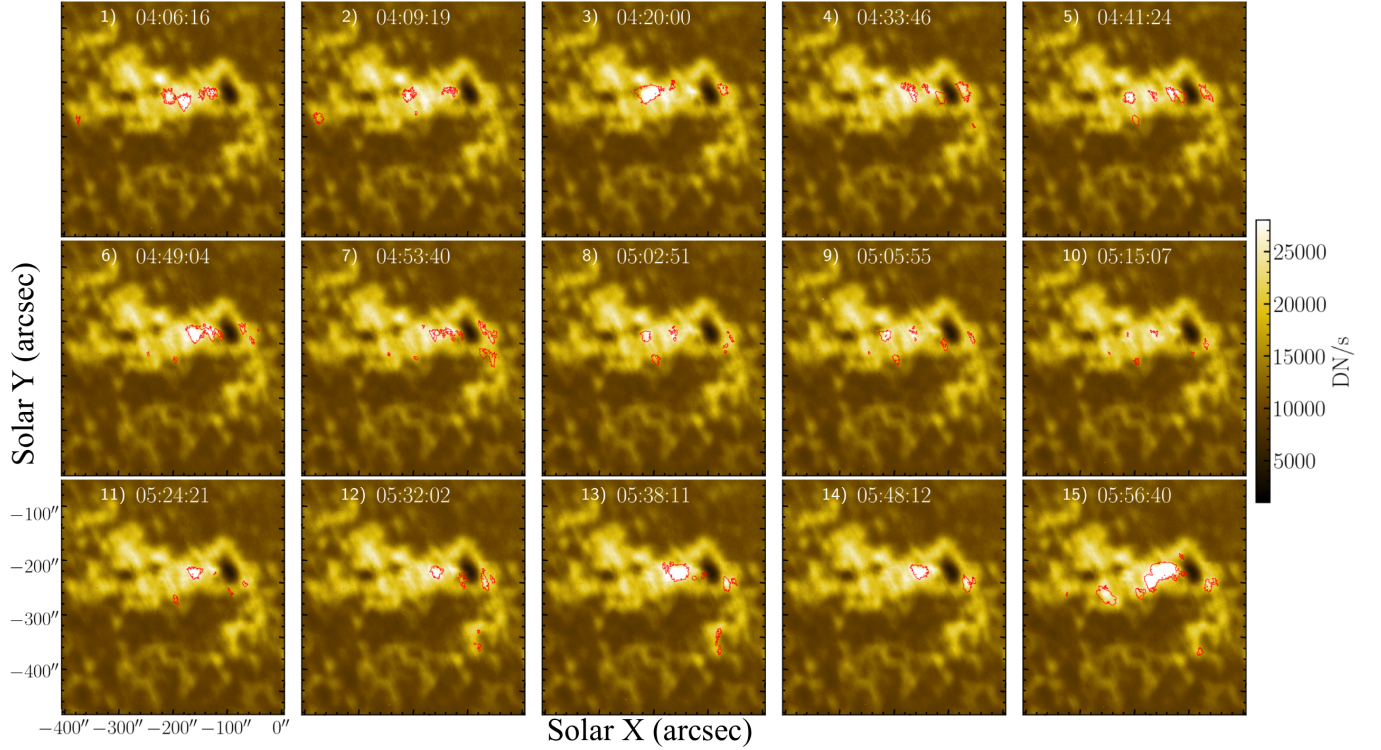


Figure A1. Case 4 timestamps

Table A2. Case 5 pre-flare timestamps and corresponding intensities active region 13738.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-07-10 13:42:07.116	126.49	-								
2	2024-07-10 13:45:46.124	36.47	-								
3	2024-07-10 13:54:53.644	52.94	-								
4	2024-07-10 14:02:11.661	18.90	-								
5	2024-07-10 14:13:08.687	61.01	-								
6	2024-07-10 14:22:16.207	78.48	15								
7	2024-07-10 14:31:23.727	39.53	10								
8	2024-07-10 14:35:02.735	42.86	-								
9	2024-07-10 14:38:41.744	45.80	-								
10	2024-07-10 14:49:38.767	25.35	-								
11	2024-07-10 14:56:56.785	31.68	-								
12	2024-07-10 15:07:53.809	73.48	-								
13	2024-07-10 15:11:32.851	64.78	-								
14	2024-07-10 15:14:20.583	54.28	-								
15	2024-07-10 15:20:49.415	52.73	-								

Table A3. Case 6 pre-flare timestamps and corresponding intensities active region 13838.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-10-08 23:59:25.593	18.37	-								
2	2024-10-09 00:02:39.513	46.59	-								
3	2024-10-09 00:15:35.583	4.88	-								
4	2024-10-09 00:20:26.848	7.15	-								
5	2024-10-09 00:26:55.200	14.93	-								
6	2024-10-09 00:36:37.761	20.45	-								
7	2024-10-09 00:41:28.961	13.67	-								
8	2024-10-09 00:46:20.342	28.18	-								
9	2024-10-09 00:52:48.196	23.51	-								
10	2024-10-09 00:56:02.275	16.86	-								
11	2024-10-09 01:11:08.484	23.48	-								
12	2024-10-09 01:15:59.492	10.32	-								

Table A4. Case 7 pre-flare timestamps and corresponding intensities active region 13838.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-11-01 00:26:43.619	23.71	-								
2	2024-11-01 00:36:54.678	17.29	-								
3	2024-11-01 00:39:53.162	12.60	-								
4	2024-11-01 00:48:49.843	9.26	-								
5	2024-11-01 00:53:18.260	6.57	-								
6	2024-11-01 00:57:46.677	11.97	-								
7	2024-11-01 01:03:43.959	18.78	-								
8	2024-11-01 01:06:42.486	17.97	-								
9	2024-11-01 01:15:38.440	4.49	-								
10	2024-11-01 01:21:35.350	9.80	-								
11	2024-11-01 01:24:33.495	11.69	-								
12	2024-11-01 01:29:01.176	21.12	-								
13	2024-11-01 01:36:26.714	9.49	-								
14	2024-11-01 01:40:53.979	23.99	-								
15	2024-11-01 01:45:21.243	34.39	17								
16	2024-11-01 01:51:17.596	18.19	-								
17	2024-11-01 01:55:44.861	45.63	69								
18	2024-11-01 02:00:12.125	11.76	-								

Table A5. Case 8 pre-flare timestamps and corresponding intensities active region 13838.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-11-01 12:40:09.863	4.73	-								
2	2024-11-01 12:44:21.873	1.87	-								
3	2024-11-01 12:48:34.739	14.60	-								
4	2024-11-01 12:51:23.314	25.41	594								
5	2024-11-01 13:02:38.345	42.54	-								
6	2024-11-01 13:13:40.179	72.29	-								
7	2024-11-01 13:19:18.484	32.38	16								
8	2024-11-01 13:26:21.273	3.15	-								
9	2024-11-01 13:31:59.699	3.66	-								
10	2024-11-01 13:36:14.132	6.65	-								
11	2024-11-01 13:41:53.429	5.46	-								
12	2024-11-01 13:46:10.198	7.31	-								
13	2024-11-01 13:51:49.528	5.75	28								
14	2024-11-01 13:56:03.958	6.76	38								
15	2024-11-01 14:00:18.358	4.94	-								
16	2024-11-01 14:07:23.735	23.24	-								
17	2024-11-01 14:14:29.112	35.18	-								

Table A6. Case 9 pre-flare timestamps and corresponding intensities active region 13838.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-11-12 22:24:40.142	41.41	-								
2	2024-11-12 22:27:54.637	36.58	-								
3	2024-11-12 22:38:10.349	2.52	-								
4	2024-11-12 22:44:39.341	3.00	-								
5	2024-11-12 22:49:31.858	8.43	-								
6	2024-11-12 22:54:22.830	51.14	10								
7	2024-11-12 23:05:43.568	21.57	-								
8	2024-11-12 23:12:12.560	7.26	-								
9	2024-11-12 23:15:27.577	7.13	-								
10	2024-11-12 23:25:10.545	99.22	-								
11	2024-11-12 23:28:25.417	108.47	18								
12	2024-11-12 23:41:23.266	20.52	-								
13	2024-11-12 23:46:14.771	24.01	-								
14	2024-11-12 23:55:58.260	59.38	-								
15	2024-11-13 00:04:04.501	36.00	20								

Table A7. Case 10 pre-flare timestamps and corresponding intensities active region 13838.

No.	Timestamp (UT)	Intensity (DN $\times 10^6$)	HEL1OS (Counts above 12 keV)	STIX	NB08 (DN)	NB01 (DN)	NB02 (DN)	NB05 (DN)	NB06 (DN)	NB07 (DN)	location
1	2024-11-13 15:11:26.400	84.69	-								
2	2024-11-13 15:21:08.385	7.81	-								
3	2024-11-13 15:24:22.402	18.32	-								
4	2024-11-13 15:27:36.384	21.02	-								
5	2024-11-13 15:34:04.258	25.76	-								
6	2024-11-13 15:45:56.358	14.53	-								
7	2024-11-13 15:49:09.956	18.18	-								
8	2024-11-13 16:08:33.636	44.99	-								
9	2024-11-13 16:26:20.267	60.71	-								
10	2024-11-13 16:31:11.239	46.00	-								
11	2024-11-13 16:39:16.134	54.26	-								
12	2024-11-13 16:48:57.896	58.41	-								

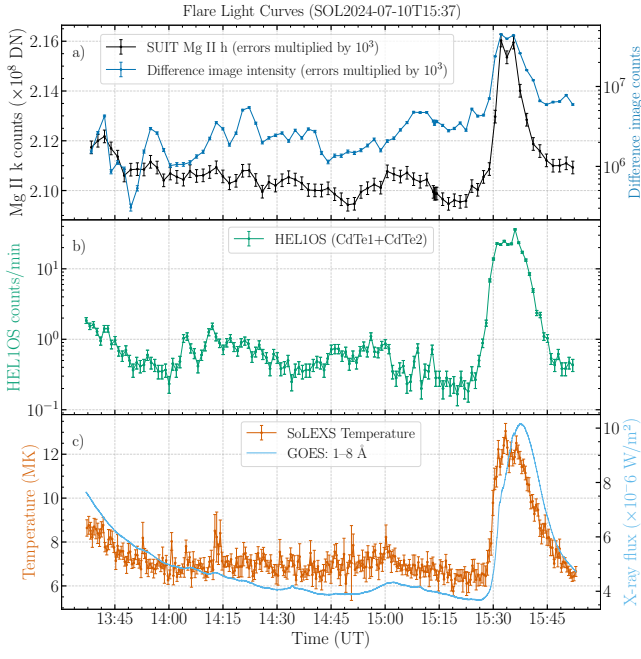


Figure C1. Case 5 Light curves

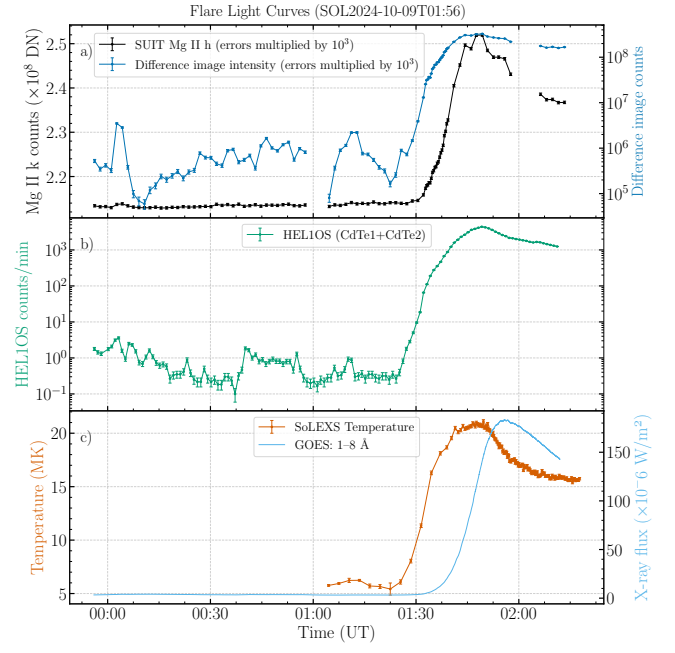


Figure C3. Case 6 Light curves

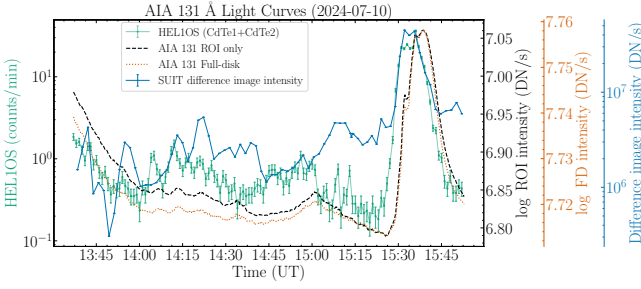


Figure C2. Case 5: AIA 131 light curve

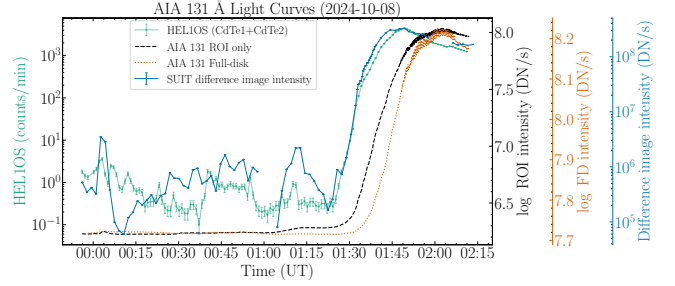


Figure C4. Case 6: AIA 131 light curve

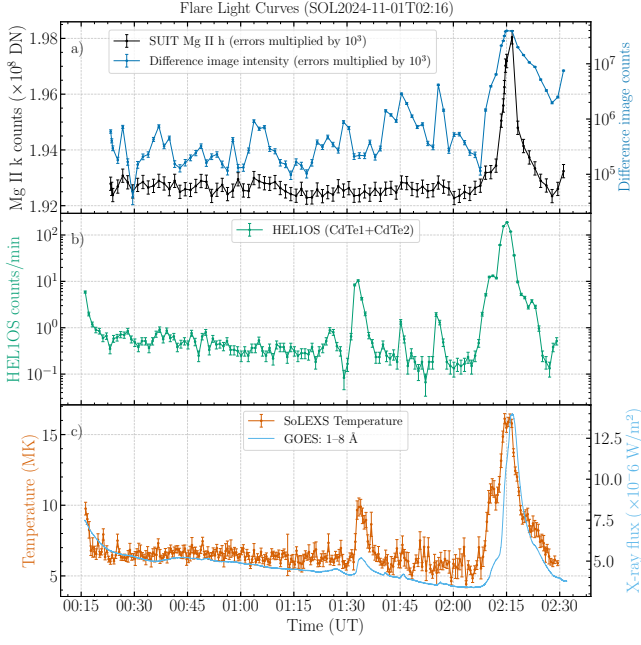


Figure C5. Case 7 Light curves

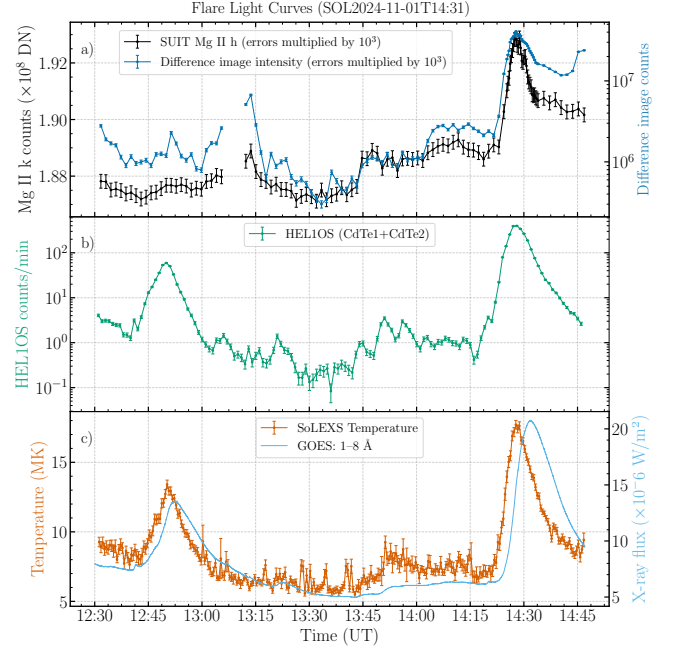


Figure C7. Case 8 Light curves

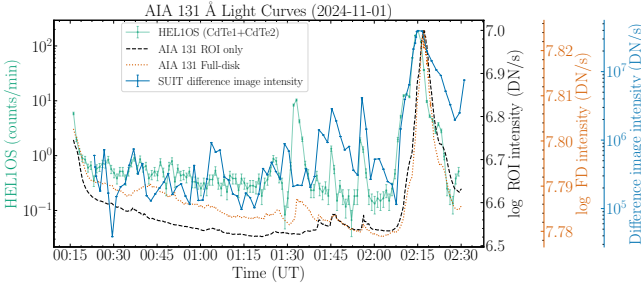


Figure C6. Case 7: AIA 131 light curve

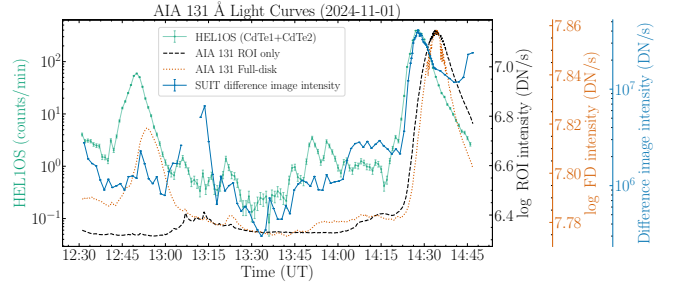


Figure C8. Case 8: AIA 131 light curve

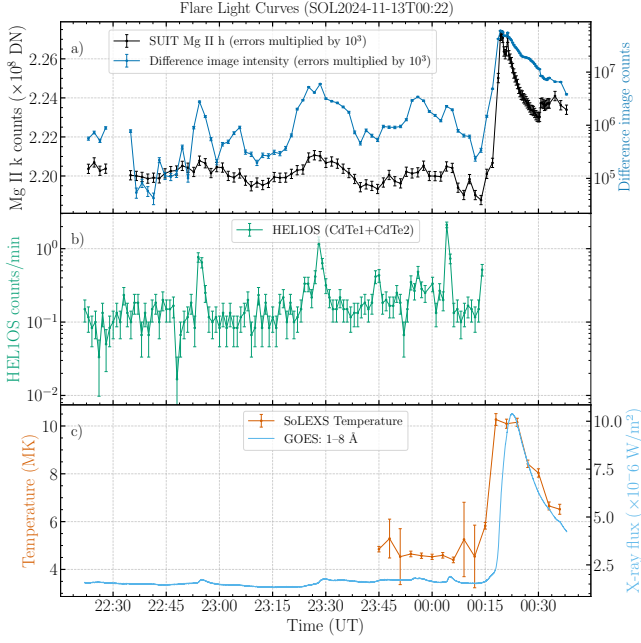


Figure C9. Case 9 Light curves

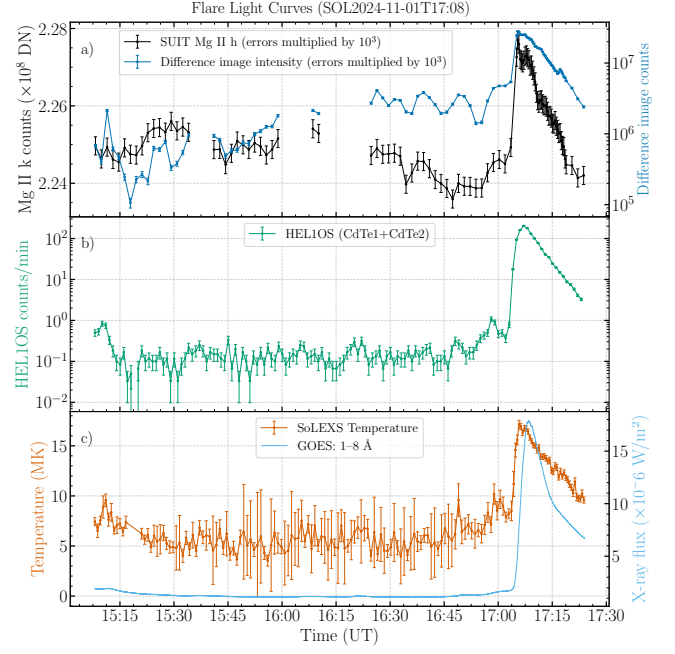


Figure C11. Case 10 Light curves

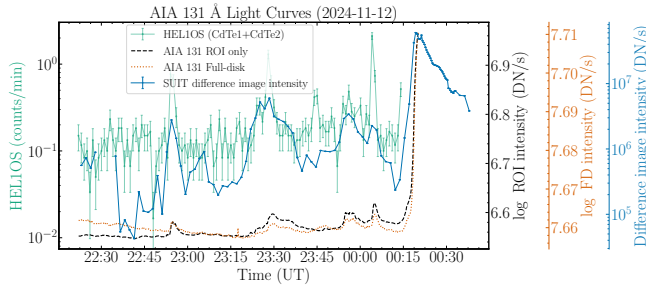


Figure C10. Case 9: AIA 131 light curve

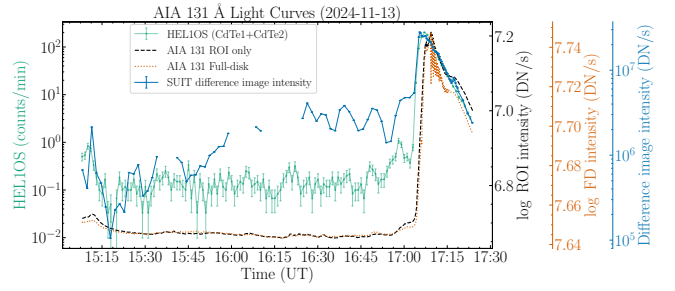


Figure C12. Case 10: AIA 131 light curve

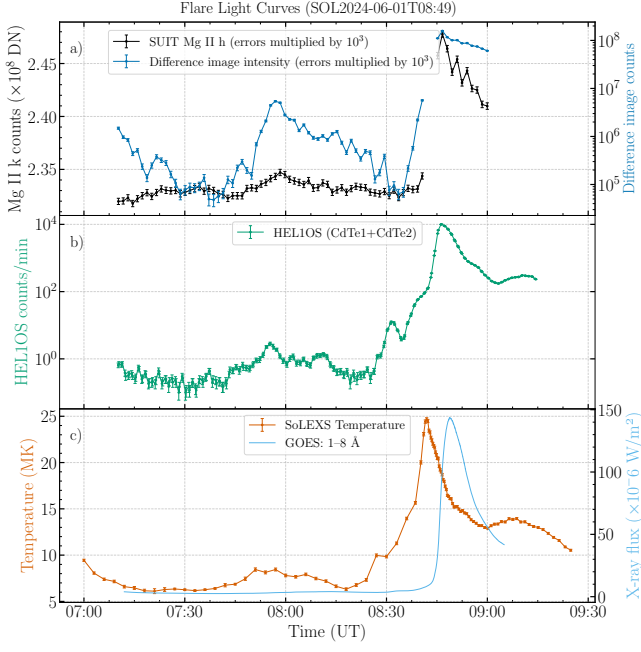


Figure C13. Case 1: June 1st flare

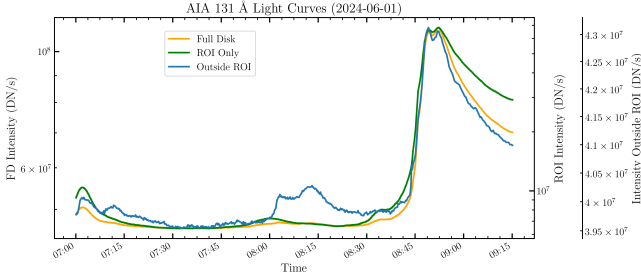


Figure C14. AIA 131 light curve

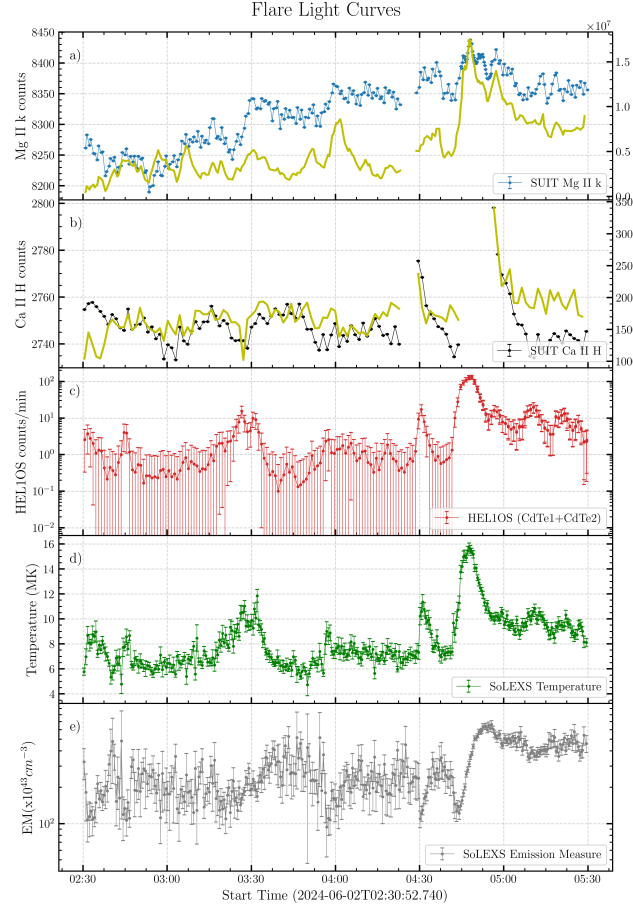


Figure C15. Case 2 light curves

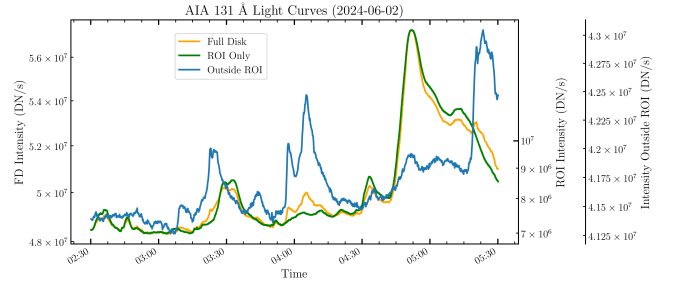
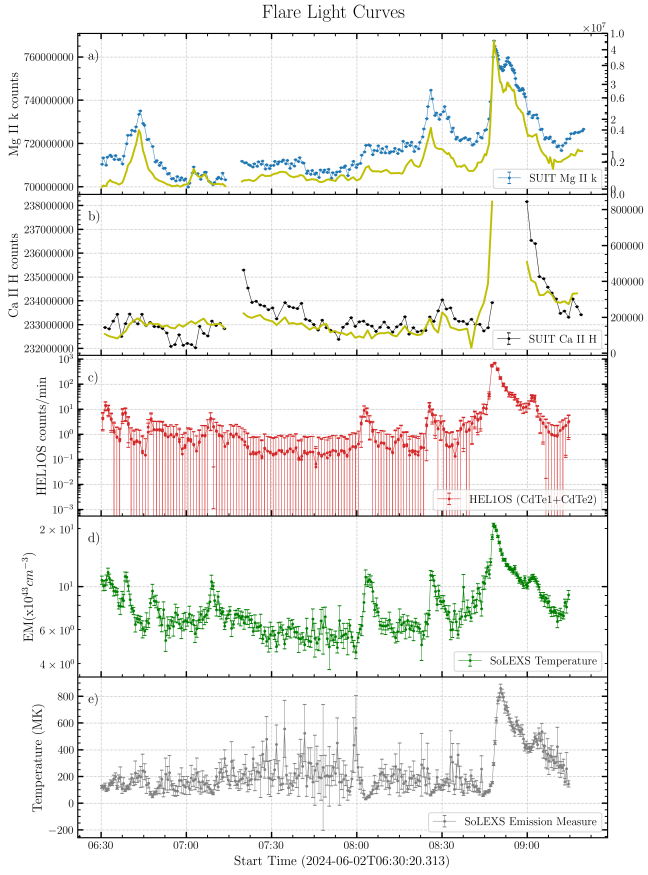
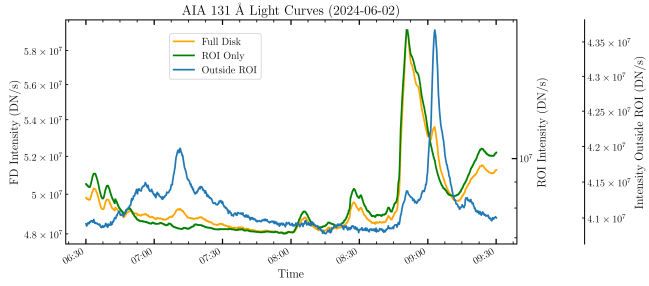


Figure C16. AIA 131 light curve

**Figure C17.** Case 3 light curves**Figure C18.** AIA 131 light curve